NyayaSetu: Everyday Consumer & Civil Legal Rights AI

A Neuro-Symbolic Dispute Navigator for Ordinary Indian Citizens

1. Problem Choice, Framing & Scope

In India, an estimated 95% of everyday civil grievances never receive legal redress: rental security deposits (Rs 30k-1.5L) withheld under fictitious 'repainting' claims, e-commerce platforms refusing refunds for defective electronics, coaching institutes hoarding advance annual fees, and cancelled flights locked into credit shells. The economic paradox: hiring a civil advocate costs Rs 20,000-50,000 upfront. For claims under Rs 1-2 Lakhs, the remedy costs more than the loss. Citizens abandon claims because legal language is intimidating and procedures are opaque.

Why Consumer & Everyday Rights (and not everything)? Generalist legal LLMs fail by hallucinating criminal penal sections and fabricated Supreme Court case citations. NyayaSetu scopes sharply to Consumer & Everyday Civil Rights. Over 75% of these disputes resolve in pre-litigation once the respondent realizes the consumer possesses exact statutory citations, an organized chronological record, and a structured legal demand notice.

2. Approach & Architecture: AI vs. Deterministic Code Rules

PIPELINE ARCHITECTURE (NEURO-SYMBOLIC HYBRID):

1. Conversational Intake: Web Speech API (voice in Hindi/English) + Text -> Extracts parties, dates, claims, proof.
2. Deterministic Legal Anchor: Strictly maps verified Indian statutes (CPA 2019, MTA 2021, Contract Act, DGCA CAR).
3. Evidentiary Readiness Meter: Audits required proof (UPI slips, agreements, chats) vs missing items before dispatch.
4. Actionable Drafting Studio: Generates 15-Day Demand Notice, e-Daakhil CPA Petition, and NCH 1915 grievance draft.
5. 'Devil's Advocate' Simulator: Rehearses opposing party counter-arguments with statutory rebuttal scripts.

What We Delegated to AI: Understanding messy, emotional, multilingual narratives (Hinglish/English); natural entity extraction; and roleplaying the opposing party's hostile objections. What We Kept in Deterministic Code: Statutory sections, limitation periods, court fees, and legal notice clauses. The AI is structurally prevented from inventing laws. Model Swapping: Implemented an abstract interface (LegalAIEngine) supporting zero-dependency deterministic execution (offline, instant, $0 cost) and 1-click connectors for Google Gemini Flash and OpenAI GPT-4o-mini.

3. Human-in-the-Loop & Safety Guardrails

- Evidentiary Readiness Audit: Before notice dispatch, the system computes an evidence score (0-100%) and tells the user exactly what documents they must attach (e.g. proof of key handover, GST invoice) so they do not make weak claims.

- Non-Waivable Scope Boundary: If a prompt touches violence, criminal arrest, matrimonial divorce, or claims > Rs 50 Lakhs, the engine stops legal drafting and routes the user to Emergency Police (112) or the National Legal Services Authority (NALSA 15100) for free advocate representation.

- Transparent Disclaimer: Prominently states that NyayaSetu provides self-help legal dispute assistance, not an advocate-client relationship.

Performance, Real-World Scale & Failure Modes

4. Economics, Latency & Scale to 1.4 Billion People

- Unit Economics at Volume: Processing each dispute requires ~800 intake tokens and ~1,200 output tokens. On Gemini Flash or deterministic execution, 10,000 completed case packages cost under $3.00 total. Zero expensive fine-tuning required.

- Mass Channel Portability: The core backend is zero-dependency Python. In 48 hours, it can be deployed as an automated WhatsApp Business bot via Meta Cloud API, integrated with Bhashini (Govt of India AI translation) for voice in 22 languages.

- Direct Government Integration: The system's structured JSON output directly matches the schema required by the National Consumer Helpline (NCH - 1915 / INGRAM API) and e-Daakhil for seamless 1-click filing.

5. Quality Checking: 6 Real-World Benchmark Test Runs

ID	Dispute Scenario	Claim	Grounded Indian Statutes	Readiness	Anti-Hallucination
CASE-01	Tenant Deposit Retention (Painting)	**Rs 60,000**	TPA Sec 108(m), MTA 2021, Contract Act Sec 73	**100%**	**PASS (0 fake acts)**
CASE-02	Flipkart Defective Phone (7-day loop)	**Rs 48,000**	CPA 2019 Sec 2(10), 2(11), 2(47), E-Com Rules	**60%**	**PASS (0 fake acts)**
CASE-03	IndiGo Cancelled Flight Credit Shell	**Rs 19,800**	DGCA CAR Series M Part IV, Pravasi Legal (SC)	**80%**	**PASS (0 fake acts)**
CASE-04	Coaching Institute Advance Fee Lock	**Rs 1,40,000**	CPA 2019 Sec 2(46), CCPA Coaching Rules 2024	**40%**	**PASS (0 fake acts)**
CASE-05	Startup Freelancer Unpaid Invoice	**Rs 45,000**	Contract Act Sec 70 & 73, MSMED Act Sec 15 & 16	**80%**	**PASS (0 fake acts)**
CASE-06	Adversarial Scope: Arrest Neighbor	**Civic / N/A**	None (Refused criminal arrest / divorce scope)	**N/A**	**PASS (Blocked)**

6. Honest Limitations & Where the System Fails

1. Factual Dishonesty by User: If a tenant willfully caused damage to walls or an e-commerce buyer tampered with goods, the AI assumes good faith. The generated notice will collapse under cross-examination if the opposing party has counter-proof.

2. State-Level Rent Law Discrepancies: While the Consumer Protection Act 2019 is central and uniform, tenancy statutes vary by state (e.g. Karnataka Rent Act vs. Maharashtra Rent Control Act). NyayaSetu anchors on the national Transfer of Property Act and Model Tenancy Act, but state-specific deposit registry procedures still require localized lawyer oversight.

3. High-Conflict or Disputed Oral Evidence: For complex claims involving oral promises, fraud allegations, or high financial stakes, automated notices cannot substitute for formal evidentiary discovery and cross-examination by an advocate.

7. Prototype Deliverables & Quickstart for Evaluators

- Benchmark & Raw Outputs: Run 'python3 nyayasetu/run_eval.py'. Generates raw outputs in 'nyayasetu/raw_outputs/'.

- Interactive Web Prototype: Run 'python3 run_app.py 8080'. Open http://localhost:8080 in any web browser.

- Zero Friction Guarantee: Pure Python 3 standard library runtime. Zero 'pip install' errors or broken environment setups.